SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I P. HARINI (192521402) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Harini

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%	1	1	1	2
Identification of Key Issues	20%	1	2	1	1
Application of Concepts	25%	2	1	2	1
Decision Making	20%	1	1	1	2
Clarity of Response	15%	1	1	1	1

25/25

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.

2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understandin	Shows good understandi ng with minor gaps	Partial understandin g; misses key aspects	Misinterpret s or fails to understand the scenario

		g of the scenario and context			
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned , logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Scenario-Based Questions

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned: School of Computing → /25, School of Electronics → /26, School of Mechanical Engineering → /27. As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

Answer:

Given: Network block = 192.168.100.0/24 (total $2^8 = 256$ addresses). The prefix lengths /25, /26 and /27 are assigned to the three departments.

Step 1: Subnet mask and block size of each prefix

Number of host bits = 32 − prefix length Block size = 2^h

Department	Prefix	Subnet Mask	Host bits (h)	Block size
School of Computing	/25	255.255.255.128	7	128
School of Electronics	/26	255.255.255.192	6	64
School of Mechanical Engg.	/27	255.255.255.224	5	32

Decision: The subnets are allocated one after the other, from the largest block to the smallest, starting at 192.168.100.0. This keeps the subnets continuous, avoids any overlap between them and leaves one single free block at the end of the network for future use.

Task 1: Network address and broadcast address Network address = first address of the block

Broadcast address = Network address + block size − 1

School of Computing: 192.168.100.0 + 128 − 1 = 192.168.100.127

School of Electronics: starts at 192.168.100.128; 128 + 64 − 1 = 191, so 192.168.100.191

School of Mechanical Engg.: starts at 192.168.100.192; 192 + 32 − 1 = 223, so 192.168.100.223

Department	Subnet	Network Address	Broadcast Address
School of Computing	192.168.100.0/25	192.168.100.0	192.168.100.127
School of Electronics	192.168.100.128/26	192.168.100.128	192.168.100.191
School of Mechanical Engg.	192.168.100.192/27	192.168.100.192	192.168.100.223

Task 2: Valid host IP address range

First host = Network address + 1 Last host = Broadcast address − 1

Department	First Usable Host	Last Usable Host
School of Computing	192.168.100.1	192.168.100.126
School of Electronics	192.168.100.129	192.168.100.190
School of Mechanical Engg.	192.168.100.193	192.168.100.222

Task 3: Number of usable host addresses

Usable hosts = $2^h - 2$ (one address is for the network and one for the broadcast)

School of Computing: $2^7 - 2 = 128 - 2 = 126$ hosts

School of Electronics: $2^6 - 2 = 64 - 2 = 62$ hosts

School of Mechanical Engg.: $2^5 - 2 = 32 - 2 = 30$ hosts

Task 4: Remaining unused address range

Addresses used = 128 + 64 + 32 = 224 (192.168.100.0 to 192.168.100.223)

Total addresses in 192.168.100.0/24 = 256

Unused addresses = 256 − 224 = 32

The unused range is 192.168.100.224 to 192.168.100.255, which is the block 192.168.100.224/27 (mask 255.255.255.224). Its network address is 192.168.100.224, its broadcast address is 192.168.100.255 and it can still provide 30 usable hosts (192.168.100.225 to 192.168.100.254) for a future department.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block 10.10.0.0/16. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks: Software Development → /24, Quality Assurance → /25, Human Resources → /26, Executive Management → /27. As the network administrator, perform the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid range of host IP addresses.
3. Calculate the number of usable hosts in each subnet.
4. Identify the unused IP address range remaining within the allocated IP block.

Answer:

Given: Network block = 10.10.0.0/16 (total 2^{16} = 65,536 addresses). The prefix lengths /24, /25, /26 and /27 are assigned to the four departments.

Step 1: Subnet mask and block size of each prefix

Department	Prefix	Subnet Mask	Host bits (h)	Block size
Software Development	/24	255.255.255.0	8	256
Quality Assurance	/25	255.255.255.128	7	128
Human Resources	/26	255.255.255.192	6	64
Executive Management	/27	255.255.255.224	5	32

Decision: The largest subnet (Software Development) is placed first at 10.10.0.0. A /24 block fills the whole third octet value 0, so the next free address is 10.10.1.0, where Quality Assurance starts. Human Resources and Executive Management follow immediately after it. This continuous allocation avoids overlap and keeps the remaining space as one large free range.

Task 1: Network address and broadcast address

Software Development: 10.10.0.0 to 10.10.0.255 (256 addresses)

Quality Assurance: starts at 10.10.1.0; 0 + 128 − 1 = 127, so 10.10.1.127

Human Resources: starts at 10.10.1.128; 128 + 64 − 1 = 191, so 10.10.1.191

Executive Management: starts at 10.10.1.192; 192 + 32 − 1 = 223, so 10.10.1.223

Department	Subnet	Network Address	Broadcast Address
Software Development	10.10.0.0/24	10.10.0.0	10.10.0.255
Quality Assurance	10.10.1.0/25	10.10.1.0	10.10.1.127
Human Resources	10.10.1.128/26	10.10.1.128	10.10.1.191
Executive Management	10.10.1.192/27	10.10.1.192	10.10.1.223

Task 2: Valid host IP address range

Department	First Usable Host	Last Usable Host
Software Development	10.10.0.1	10.10.0.254
Quality Assurance	10.10.1.1	10.10.1.126
Human Resources	10.10.1.129	10.10.1.190
Executive Management	10.10.1.193	10.10.1.222

Task 3: Number of usable hosts

Usable hosts = $2^h - 2$

Software Development: $2^8 - 2 = 256 - 2 = 254$ hosts

Quality Assurance: $2^7 - 2 = 128 - 2 = 126$ hosts

Human Resources: $2^6 - 2 = 64 - 2 = 62$ hosts

Executive Management: $2^5 - 2 = 32 - 2 = 30$ hosts

Task 4: Remaining unused address range

Addresses used = 256 + 128 + 64 + 32 = 480 (10.10.0.0 to 10.10.1.223)

Total addresses in 10.10.0.0/16 = 65,536

Unused addresses = 65,536 − 480 = 65,056

The unused range is 10.10.1.224 to 10.10.255.255. It begins with the free block 10.10.1.224/27 and continues from 10.10.2.0 up to 10.10.255.255, so it is available for future departments and expansion.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network 2001:0db8:abcd:0000:0000:0000:0000:0000/64. As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its compressed notation.
2. Expand the compressed address back into its full hexadecimal representation.
3. Identify the network prefix using the /64 prefix length.
4. Calculate the total number of host addresses supported within the subnet.
5. Find the total number of possible host addresses available within this network.

Answer:

Given: IPv6 network = 2001:0db8:abcd:0000:0000:0000:0000:0000/64. An IPv6 address has 128 bits, written as 8 groups of 4 hexadecimal digits.

Task 1: Compressed notation

Rule 1: Remove the leading zeros in every group.

2001:0db8:abcd:0000:0000:0000:0000:0000 → 2001:db8:abcd:0:0:0:0:0

Rule 2: Replace the longest run of consecutive all-zero groups with :: (only once).

2001:db8:abcd:0:0:0:0:0 → 2001:db8:abcd::

Compressed form = 2001:db8:abcd::/64

Task 2: Expanding the compressed address

Given compressed address = 2001:db8:abcd::

Groups written = 3 (2001, db8, abcd). Groups hidden by :: = 8 − 3 = 5 groups of zeros.

Add the 5 zero groups and pad every group to 4 digits (db8 becomes 0db8):

Full form = 2001:0db8:abcd:0000:0000:0000:0000:0000/64

Task 3: Network prefix

Prefix length /64 means the first 64 bits are the network part and the last 64 bits are the host part.

64 bits = 4 groups of 16 bits = 2001 : 0db8 : abcd : 0000

Global Routing Prefix	Subnet ID	Interface ID (Host part)
2001:0db8:abcd	0000	0000:0000:0000:0000
48 bits	16 bits	64 bits

Network prefix = 2001:db8:abcd:0::/64 (written as 2001:db8:abcd::/64)

Task 4: Host addresses supported in the subnet

Host bits = 128 − 64 = 64

Number of host addresses = 2^{64} = 18,446,744,073,709,551,616

IPv6 has no broadcast address, so the "− 2" used in IPv4 is not applied. Strictly, the all-zeros interface ID is reserved as the Subnet-Router anycast address, so $2^{64} - 1$ addresses can be given to devices, which is still about 1.8×10^{19} addresses.

Task 5: Total possible host addresses in this network

Lowest address = 2001:db8:abcd:0:0:0:0:0

Highest address = 2001:db8:abcd:0:ffff:ffff:ffff:ffff

Total addresses = 2^{64} = 18,446,744,073,709,551,616

This number is far more than the hospital needs for its medical devices and IoT sensors, so address shortage will not occur in this subnet.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries: 192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24. A packet arrives at the router with the destination IP address 192.168.2.45. As the routing engineer, determine:

1. The destination subnet to which the packet belongs.
2. The matching routing table entry.
3. The next-hop network/interface used to forward the packet.
4. The default route that should be used if the destination does not match any routing table entry.

Answer:

Given: Destination IP address = 192.168.2.45. Every routing table entry has the prefix /24, so the subnet mask is 255.255.255.0.

Task 1: Destination subnet

The router performs a bitwise AND of the destination address with the subnet mask.

192.168.2.45 = 11000000.10101000.00000010.00101101

255.255.255.0 = 11111111.11111111.11111111.00000000

AND result = 11000000.10101000.00000010.00000000 = 192.168.2.0

The destination subnet is 192.168.2.0/24. Its network address is 192.168.2.0, its broadcast address is 192.168.2.255 and its host range is 192.168.2.1 to 192.168.2.254, so 192.168.2.45 is a valid host address in this subnet.

Task 2: Matching routing table entry

The destination address is compared with each entry of the routing table:

Routing Entry	Address Range	Contains 192.168.2.45?
192.168.1.0/24	192.168.1.0 – 192.168.1.255	No
192.168.2.0/24	192.168.2.0 – 192.168.2.255	Yes
192.168.3.0/24	192.168.3.0 – 192.168.3.255	No

Only the entry 192.168.2.0/24 matches. Therefore the matching routing table entry is 192.168.2.0/24 (longest prefix match, /24).

Task 3: Next hop / exit interface

The router forwards the packet using the next hop or exit interface that is stored against the matching entry 192.168.2.0/24, that is, the interface connected to the customer network 192.168.2.0/24. The question does not give interface names, so the interfaces below are assumed only to show the forwarding clearly:

Destination Network	Exit Interface (assumed)	Used for 192.168.2.45?
192.168.1.0/24	GigabitEthernet0/0	No
192.168.2.0/24	GigabitEthernet0/1	Yes
192.168.3.0/24	GigabitEthernet0/2	No

The packet is therefore forwarded through GigabitEthernet0/1 towards the customer network 192.168.2.0/24, and the router delivers it directly to the host 192.168.2.45.

Task 4: Default route

If the destination does not match any entry in the routing table, the router uses the default route 0.0.0.0/0. It matches every address and sends the packet to the next-hop router of the ISP towards the Internet. It is configured as:

Router(config)# ip route 0.0.0.0 0.0.0.0 <next-hop IP address or exit interface>

For example, a packet for 192.168.4.10 matches none of the three entries, so it is forwarded by the default route.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows: Administration LAN → 192.168.10.0/26 and Finance LAN →

192.168.10.64/26. The router interfaces are configured with: Administration Gateway → 192.168.10.1 and Finance Gateway → 192.168.10.65. As the network administrator, perform the following tasks:

1. Determine the network address and broadcast address for each LAN.
2. Identify the valid host IP address range.
3. Suggest suitable IP addresses for three computers in each LAN.
4. Identify the default gateway that hosts in each LAN should use for communication.

Answer:

Given: Administration LAN = 192.168.10.0/26, Finance LAN = 192.168.10.64/26.

Prefix /26 → subnet mask = 255.255.255.192

Host bits = 32 − 26 = 6 Block size = 2^6 = 64

Usable hosts in each LAN = $2^6 - 2 = 62$

Task 1: Network address and broadcast address

Administration LAN: network = 192.168.10.0; broadcast = 0 + 64 − 1 = 63, so 192.168.10.63

Finance LAN: network = 192.168.10.64; broadcast = 64 + 64 − 1 = 127, so 192.168.10.127

LAN	Subnet	Network Address	Broadcast Address
Administration LAN	192.168.10.0/26	192.168.10.0	192.168.10.63
Finance LAN	192.168.10.64/26	192.168.10.64	192.168.10.127

Task 2: Valid host IP address range

LAN	First Usable Host	Last Usable Host
Administration LAN	192.168.10.1	192.168.10.62
Finance LAN	192.168.10.65	192.168.10.126

Task 3: IP addresses for three computers in each LAN

The first address of each LAN is used by the router gateway, so the computers are given the next free addresses from the valid host range.

LAN	Computer	IP Address	Subnet Mask	Default Gateway
Administration	PC1	192.168.10.2	255.255.255.192	192.168.10.1
Administration	PC2	192.168.10.3	255.255.255.192	192.168.10.1
Administration	PC3	192.168.10.4	255.255.255.192	192.168.10.1
Finance	PC1	192.168.10.66	255.255.255.192	192.168.10.65
Finance	PC2	192.168.10.67	255.255.255.192	192.168.10.65
Finance	PC3	192.168.10.68	255.255.255.192	192.168.10.65

Task 4: Default gateway

Administration LAN hosts → default gateway = 192.168.10.1

Finance LAN hosts → default gateway = 192.168.10.65

Justification: A default gateway must belong to the same subnet as the host, because a host can reach it directly without any routing. The address 192.168.10.1 lies inside 192.168.10.0/26 and 192.168.10.65 lies inside 192.168.10.64/26. When a computer wants to communicate with a computer of the other LAN, it sends the packet to its own gateway, and the Layer-3 router forwards it to the other LAN.

RUBRICS FOR CALCULATION

Scenario Based Assessment

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand